

# TUGAS MANDIRI PEKAN 12

Nama : Muhammad Shiddiq

NIM : 0110222199

## ✓ Import Library

```
1 # Import library dasar
2 import numpy as np
3 import pandas as pd
4 from sklearn.datasets import load_wine
5 from sklearn.model_selection import train_test_split
6 from sklearn.preprocessing import StandardScaler
7 from sklearn.decomposition import PCA
8 from sklearn.svm import SVC
9 from sklearn.metrics import accuracy_score, classification_report, confusion_m
10 import matplotlib.pyplot as plt
11 from mpl_toolkits.mplot3d import Axes3D # untuk 3D plot
12 #Agar plot tampil rapi
13 plt.rcParams['figure.figsize'] = (8, 6)
```

## ✓ Load dataset

```
1 from google.colab import drive
2 drive.mount('/content/drive')
```

Mounted at /content/drive

```
1 path = '/content/drive/MyDrive/MACHINELEARNING/PRAKTIKUM/PRAKTIKUM12/DATA/'
```

```
1 df = pd.read_csv(path + "data.csv")
2 df
```

|     | id       | diagnosis | radius_mean | texture_mean | perimeter_mean | area_mean | smoothness_me |
|-----|----------|-----------|-------------|--------------|----------------|-----------|---------------|
| 0   | 842302   | M         | 17.99       | 10.38        | 122.80         | 1001.0    | 0.118         |
| 1   | 842517   | M         | 20.57       | 17.77        | 132.90         | 1326.0    | 0.084         |
| 2   | 84300903 | M         | 19.69       | 21.25        | 130.00         | 1203.0    | 0.109         |
| 3   | 84348301 | M         | 11.42       | 20.38        | 77.58          | 386.1     | 0.142         |
| 4   | 84358402 | M         | 20.29       | 14.34        | 135.10         | 1297.0    | 0.100         |
| ... | ...      | ...       | ...         | ...          | ...            | ...       | ...           |
| 564 | 926424   | M         | 21.56       | 22.39        | 142.00         | 1479.0    | 0.111         |
| 565 | 926682   | M         | 20.13       | 28.25        | 131.20         | 1261.0    | 0.097         |
| 566 | 926954   | M         | 16.60       | 28.08        | 108.30         | 858.1     | 0.084         |
| 567 | 927241   | M         | 20.60       | 29.33        | 140.10         | 1265.0    | 0.117         |
| 568 | 92751    | B         | 7.76        | 24.54        | 47.92          | 181.0     | 0.052         |

569 rows × 33 columns

## ✓ Data Preprocessing

```
1 df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 569 entries, 0 to 568
Data columns (total 33 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   id                                    569 non-null    int64
1   diagnosis                            569 non-null    object
2   radius_mean                          569 non-null    float64
3   texture_mean                         569 non-null    float64
4   perimeter_mean                       569 non-null    float64
5   area_mean                           569 non-null    float64
6   smoothness_mean                      569 non-null    float64
7   compactness_mean                    569 non-null    float64
8   concavity_mean                      569 non-null    float64
9   concave points_mean                 569 non-null    float64
10  symmetry_mean                       569 non-null    float64
11  fractal_dimension_mean              569 non-null    float64
12  radius_se                           569 non-null    float64
13  texture_se                           569 non-null    float64
14  perimeter_se                         569 non-null    float64
15  area_se                             569 non-null    float64
16  smoothness_se                       569 non-null    float64
17  compactness_se                      569 non-null    float64
18  concavity_se                        569 non-null    float64
19  concave points_se                   569 non-null    float64
20  symmetry_se                         569 non-null    float64
21  fractal_dimension_se                569 non-null    float64
22  radius_worst                        569 non-null    float64
23  texture_worst                       569 non-null    float64
24  perimeter_worst                     569 non-null    float64
25  area_worst                          569 non-null    float64
26  smoothness_worst                    569 non-null    float64
27  compactness_worst                   569 non-null    float64
28  concavity_worst                     569 non-null    float64
29  concave points_worst                569 non-null    float64
```

```
30 symmetry_worst      569 non-null    float64
31 fractal_dimension_worst  569 non-null    float64
32 Unnamed: 32          0 non-null      float64
dtypes: float64(31), int64(1), object(1)
memory usage: 146.8+ KB
```

```
1 df.isnull().sum()
```

|                         | 0   |
|-------------------------|-----|
| id                      | 0   |
| diagnosis               | 0   |
| radius_mean             | 0   |
| texture_mean            | 0   |
| perimeter_mean          | 0   |
| area_mean               | 0   |
| smoothness_mean         | 0   |
| compactness_mean        | 0   |
| concavity_mean          | 0   |
| concave points_mean     | 0   |
| symmetry_mean           | 0   |
| fractal_dimension_mean  | 0   |
| radius_se               | 0   |
| texture_se              | 0   |
| perimeter_se            | 0   |
| area_se                 | 0   |
| smoothness_se           | 0   |
| compactness_se          | 0   |
| concavity_se            | 0   |
| concave points_se       | 0   |
| symmetry_se             | 0   |
| fractal_dimension_se    | 0   |
| radius_worst            | 0   |
| texture_worst           | 0   |
| perimeter_worst         | 0   |
| area_worst              | 0   |
| smoothness_worst        | 0   |
| compactness_worst       | 0   |
| concavity_worst         | 0   |
| concave points_worst    | 0   |
| symmetry_worst          | 0   |
| fractal_dimension_worst | 0   |
| Unnamed: 32             | 569 |

**dtype:** int64

1 df.describe()

|              | id           | radius_mean | texture_mean | perimeter_mean | area_mean   | smoothness_mean |
|--------------|--------------|-------------|--------------|----------------|-------------|-----------------|
| <b>count</b> | 5.690000e+02 | 569.000000  | 569.000000   | 569.000000     | 569.000000  | 569.000000      |
| <b>mean</b>  | 3.037183e+07 | 14.127292   | 19.289649    | 91.969033      | 654.889104  | 0.096360        |
| <b>std</b>   | 1.250206e+08 | 3.524049    | 4.301036     | 24.298981      | 351.914129  | 0.014064        |
| <b>min</b>   | 8.670000e+03 | 6.981000    | 9.710000     | 43.790000      | 143.500000  | 0.052630        |
| <b>25%</b>   | 8.692180e+05 | 11.700000   | 16.170000    | 75.170000      | 420.300000  | 0.086370        |
| <b>50%</b>   | 9.060240e+05 | 13.370000   | 18.840000    | 86.240000      | 551.100000  | 0.095870        |
| <b>75%</b>   | 8.813129e+06 | 15.780000   | 21.800000    | 104.100000     | 782.700000  | 0.105300        |
| <b>max</b>   | 9.113205e+08 | 28.110000   | 39.280000    | 188.500000     | 2501.000000 | 0.163400        |

8 rows × 32 columns

```
1 df.duplicated().sum()
```

```
np.int64(0)
```

```
1 # Definisikan fitur (X) dan label (y)
2 # Drop kolom 'id', 'diagnosis', dan 'Unnamed: 32' dari fitur
3 X = df.drop(columns=['id', 'diagnosis', 'Unnamed: 32']).values
4
5 # Label: konversi 'M' ke 1 dan 'B' ke 0
6 y = df['diagnosis'].map({'M': 1, 'B': 0}).values
7 # Nama kelas untuk laporan evaluasi
8 target_names = ['Benign (0)', 'Malignant (1)']
9
10 print(f"Shape X (Fitur): {X.shape}")
11 print(f"Shape y (Label): {y.shape}")
```

Shape X (Fitur): (569, 30)  
Shape y (Label): (569,)

```
1 # Train-test split (80% train, 20% test)
2 X_train, X_test, y_train, y_test = train_test_split(
3     X, y, test_size=0.2, random_state=42, stratify=y
4 )
5
6 print(f"Shape X_train: {X_train.shape}")
7 print(f"Shape X_test: {X_test.shape}")
```

Shape X\_train: (455, 30)  
Shape X\_test: (114, 30)

```
1 # Standardisasi (mean=0, std=1)
2 scaler = StandardScaler()
3 X_train_scaled = scaler.fit_transform(X_train)
4 X_test_scaled = scaler.transform(X_test)
5
6 print("\nContoh data terstandarisasi (5 baris pertama X_train_scaled):")
7 print(X_train_scaled[:5])
```

Contoh data terstandarisasi (5 baris pertama X\_train\_scaled):  
[[ 5.18558727e-01 8.91825791e-01 4.24631702e-01 3.83925436e-01  
 -9.74743706e-01 -6.89771505e-01 -6.88586446e-01 -3.98175254e-01  
 -1.03915470e+00 -8.25056321e-01 -1.09317755e-01 -5.59755400e-02

```

-2.10096206e-01 -1.59132582e-02 -1.00518399e+00 -9.11941990e-01
-6.62815884e-01 -6.52561081e-01 -7.01889114e-01 -2.75393571e-01
 5.79797697e-01 1.31324246e+00 4.66908134e-01 4.45982711e-01
-5.96154777e-01 -6.34722227e-01 -6.10227299e-01 -2.35743918e-01
 5.45663235e-02 2.18367276e-02]
[-5.16364088e-01 -1.63971029e+00 -5.41348716e-01 -5.42961327e-01
 4.76219058e-01 -6.31833818e-01 -6.04281166e-01 -3.03074908e-01
 5.21543093e-01 -4.54522896e-01 -6.04377961e-01 -1.00104604e+00
-5.85429002e-01 -4.93453793e-01 4.03212009e-01 -7.68173276e-01
-4.79187222e-01 1.14508478e-01 -1.42950761e-01 -5.77397732e-01
-5.82458953e-01 -1.69029101e+00 -6.11934288e-01 -5.87013537e-01
 2.73581959e-01 -8.14844486e-01 -7.12666415e-01 -3.23207881e-01
-1.37576237e-01 -9.04401643e-01]
[-3.68118388e-01 4.55514964e-01 -3.88249933e-01 -4.02970039e-01
-1.43297929e+00 -3.83927370e-01 -3.42175070e-01 -7.65459348e-01
-8.50857052e-01 -2.26170901e-01 3.03979894e-01 1.05150064e+00
-1.69545176e-01 -8.09073190e-04 -3.10104091e-01 1.10632991e+00
 6.22584752e-01 2.73684564e-01 7.54482587e-01 1.50810455e+00
-3.98622224e-01 1.81977388e-01 -4.75431283e-01 -4.20778328e-01
-1.62278520e+00 -3.91399175e-01 -4.31312898e-01 -8.90825037e-01
-6.75892999e-01 -1.44015592e-01]
[ 2.05284794e-01 7.26167669e-01 4.00330308e-01 7.06116009e-02
 2.43253484e-01 2.20358457e+00 2.25609405e+00 1.21323326e+00
 8.18473995e-01 8.99791136e-01 -5.45729713e-01 -6.21676545e-01
 2.61427407e-01 -3.53584902e-01 2.44600975e-02 2.09072809e+00
 1.49056147e+00 1.69512701e+00 -6.54909382e-01 7.67547750e-01
-3.09312591e-04 2.74191135e-01 5.13776118e-01 -9.94819409e-02
 4.18538082e-01 2.86596968e+00 2.95861933e+00 1.97706439e+00
-7.56459905e-02 1.72884831e+00]
[ 1.24300470e+00 1.94195111e-01 1.21037678e+00 1.20665201e+00
-1.11441850e-01 5.13480707e-02 7.32962322e-01 7.13767250e-01
-4.27187350e-01 -8.22183969e-01 1.52386344e+00 1.14394136e+00
 1.28274755e+00 1.08131268e+00 1.35184527e+00 1.07730938e-01
 5.92926514e-01 1.18098825e+00 3.01549783e-01 1.71526944e-01
 1.01283532e+00 2.23144239e-01 9.38517229e-01 8.80910405e-01
 7.32014364e-02 -2.77005741e-01 3.27775115e-01 5.01858851e-01
-9.09322392e-01 -5.46248793e-01]]

```

## Model SVM tanpa PCA

◆ Gemini

```

1 1 # Model SVM tanpa PCA
2 2 svm_no_pca = SVC(kernel='rbf', gamma='scale', random_state=42)
3 3 svm_no_pca.fit(X_train_scaled, y_train)
4 4
5 5 # Prediksi dan evaluasi
6 6 y_pred_no_pca = svm_no_pca.predict(X_test_scaled)
7 7 acc_no_pca = accuracy_score(y_test, y_pred_no_pca)
8 8
9 9 print(f"\nAkurasi SVM tanpa PCA: {acc_no_pca:.4f}")
10 10 print("\nClassification Report (tanpa PCA):")
11 11 print(classification_report(y_test, y_pred_no_pca, target_names=target_names))

```

Akurasi SVM tanpa PCA: 0.9737

Classification Report (tanpa PCA):

|               | precision | recall | f1-score | support |
|---------------|-----------|--------|----------|---------|
| Benign (0)    | 0.96      | 1.00   | 0.98     | 72      |
| Malignant (1) | 1.00      | 0.93   | 0.96     | 42      |
| accuracy      |           |        | 0.97     | 114     |
| macro avg     | 0.98      | 0.96   | 0.97     | 114     |
| weighted avg  | 0.97      | 0.97   | 0.97     | 114     |

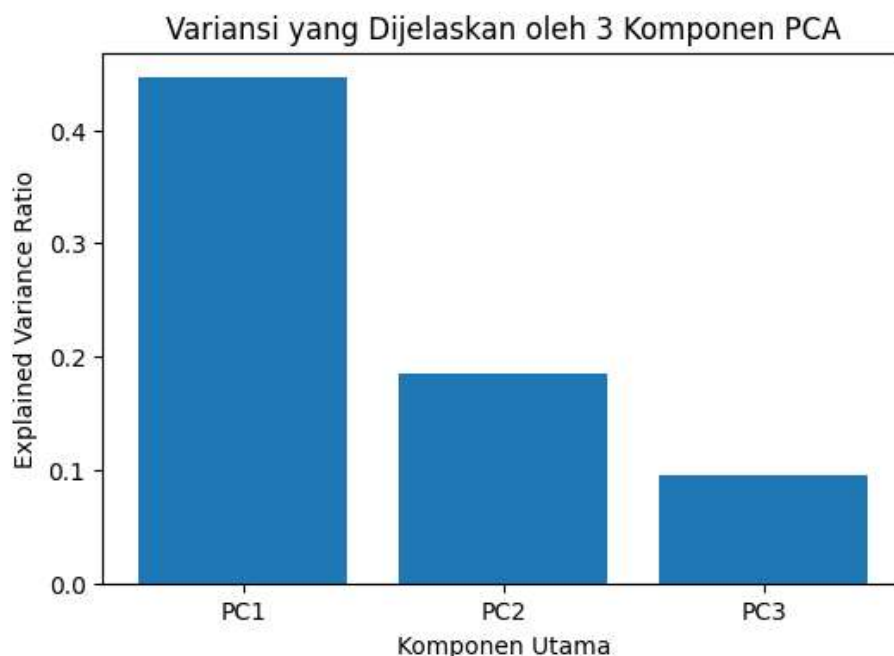
## ✓ Analisis komponen utama PCA

```
1 # PCA dengan 3 komponen utama
2 pca = PCA(n_components=3)
3 X_train_pca = pca.fit_transform(X_train_scaled)
4 X_test_pca = pca.transform(X_test_scaled)
5
6 print(f"Shape X_train_pca: {X_train_pca.shape}")
7 print(f"Shape X_test_pca: {X_test_pca.shape}")
```

```
Shape X_train_pca: (455, 3)
Shape X_test_pca: (114, 3)
```

```
1 # Variansi yang dijelaskan oleh tiap komponen
2 explained_var = pca.explained_variance_ratio_
3 print("\nExplained Variance Ratio tiap komponen:")
4 print(explained_var)
5 print(f"Total variansi yang dijelaskan 3 komponen pertama: {explained_var.sum():.4f}")
6
7 # Visualisasi Explained Variance Ratio
8 plt.figure(figsize=(6, 4))
9 plt.bar([f'PC{i+1}' for i in range(len(explained_var))], explained_var)
10 plt.xlabel('Komponen Utama')
11 plt.ylabel('Explained Variance Ratio')
12 plt.title('Variansi yang Dijelaskan oleh 3 Komponen PCA')
13 plt.show()
```

```
Explained Variance Ratio tiap komponen:
[0.44593522 0.18545255 0.09584641]
Total variansi yang dijelaskan 3 komponen pertama: 0.7272
```



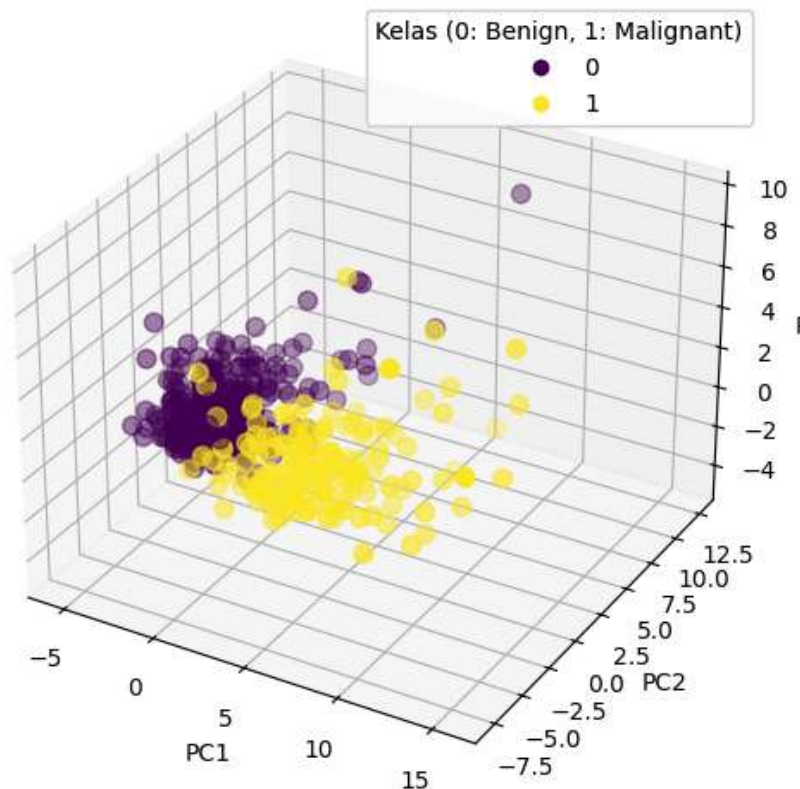
```
1 # Visualisasi 3D PCA (menggunakan data train)
2 fig = plt.figure(figsize=(8, 6))
3 ax = fig.add_subplot(111, projection='3d')
4
5 # Data untuk visualisasi
6 x_data = X_train_pca[:, 0]
7 y_data = X_train_pca[:, 1]
```

```

8 z_data = X_train_pca[:, 2]
9
10 scatter = ax.scatter(
11     x_data,
12     y_data,
13     z_data,
14     c=y_train,
15     cmap='viridis',
16     s=60
17 )
18
19 ax.set_title('Visualisasi PCA (3 Komponen) - Dataset Breast Cancer')
20 ax.set_xlabel('PC1')
21 ax.set_ylabel('PC2')
22 ax.set_zlabel('PC3')
23
24 # Tambahkan legend berdasarkan kelas
25 legend1 = ax.legend(
26     *scatter.legend_elements(),
27     title="Kelas (0: Benign, 1: Malignant)"
28 )
29 ax.add_artist(legend1)
30
31 plt.show()

```

Visualisasi PCA (3 Komponen) - Dataset Breast Cancer



## ✓ Model SVM dengan PCA

```

1 # Model SVM dengan PCA
2 svm_pca = SVC(kernel='rbf', gamma='scale', random_state=42)
3 svm_pca.fit(X_train_pca, y_train)
4
5 # Prediksi dan evaluasi
6 y_pred_pca = svm_pca.predict(X_test_pca)

```

```
7 acc_pca = accuracy_score(y_test, y_pred_pca)
8
9 print(f"\nAkurasi SVM dengan PCA (3 komponen): {acc_pca:.4f}")
10 print("\nClassification Report (dengan PCA):")
```

Akurasi SVM dengan PCA (3 komponen): 0.9561

Classification Report (dengan PCA):

|               | precision | recall | f1-score | support |
|---------------|-----------|--------|----------|---------|
| Benign (0)    | 0.94      | 1.00   | 0.97     | 72      |
| Malignant (1) | 1.00      | 0.88   | 0.94     | 42      |
| accuracy      |           |        | 0.96     | 114     |
| macro avg     | 0.97      | 0.94   | 0.95     | 114     |
| weighted avg  | 0.96      | 0.96   | 0.96     | 114     |